--- control timing results ---

cnt	avg_time	avg_send_msg_t	first_time	min	max	dev	median	correct_authorise	correct_register_value_change_%
30	0.6045262019	0.6029129028	0.7286071777	0.447511673	1.16109848	0.161739717	0.5431175232	1	1

--- stage timing averages (ms) (k,n)=(3,5) ---

stage	avg_ms
make_message	0.02863407135
total_auth_time	1.671902339
share_collect_tim	0.3776947657
key_construction	0.02058347066
message_signatu	0.17897288
zeroise_time	0.001875559489
auth_time	1.079201698
share_cost	0.5783557892
send_message	0.6558338801
total_time	2.365255356
min	1.040697098
max	27.68707275
dev	4.785330253
median	1.496553421
first_time	27.68707275

--- scenario results (k,n)=(3,5)---

scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	30	2.365255356	1	1
replay_message	30	0.8390903473	1	1
wrong_target	30	0.6147782008	1	1
mismatched_valu	30	1.504516602	1	1
outdated_ts	30	0.9429057439	1	1
missing_shares	30	0.1337448756	1	1
authorisation_ski	30	0.5912939707	1	1

--- stage timing averages (ms) (k,n)=(3,9) ---

stage	avg_ms
make_message	0.03388722738
total_auth_time	2.276539803
share_collect_tim	0.5026817322
key_construction	0.02398490906
message_signatu	0.2068916957
zeroise_time	0.002272923787
auth_time	1.52156353
share_cost	0.7351239522
send_message	0.9890317917
total_time	3.313175837
min	1.282930374
max	35.94779968
dev	6.278436768
median	1.611590385
first_time	35.94779968

--- scenario results (k,n)=(3,9)---

scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	30	3.313175837	1	1
replay_message	30	1.143980026	1	1
wrong_target	30	0.761961937	1	1
mismatched_valu	30	2.093362808	1	1
outdated_ts	30	1.047889392	1	1
missing_shares	30	0.2086321513	1	1
authorisation_ski	30	0.7887045542	1	1

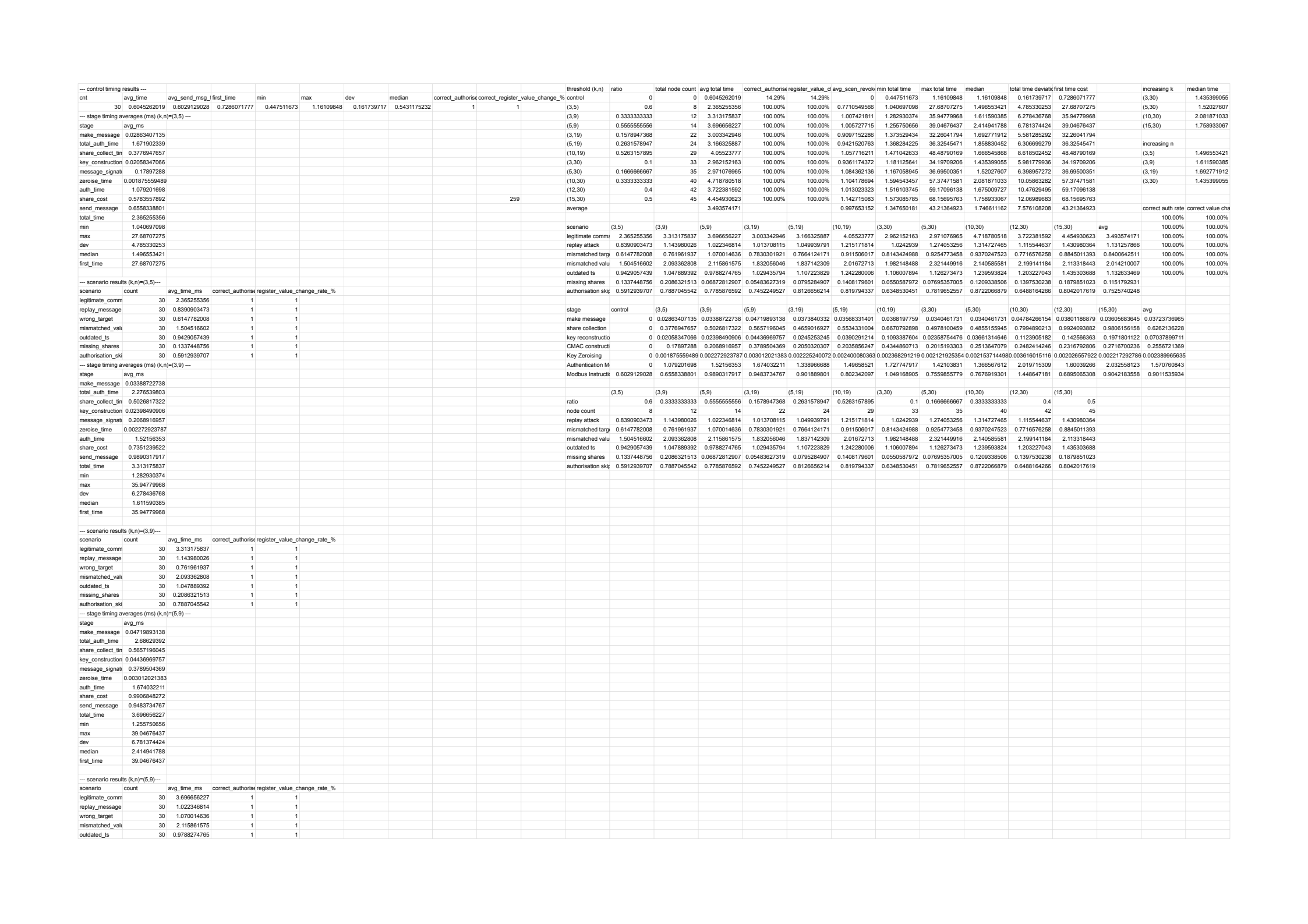
--- stage timing averages (ms) (k,n)=(5,9) ---

stage	avg_ms
make_message	0.04719893138
total_auth_time	2.68629392
share_collect_tim	0.5657196045
key_construction	0.04436969757
message_signatu	0.3789504369
zeroise_time	0.003012021383
auth_time	1.674032211
share_cost	0.9906848272
send_message	0.9483734767
total_time	3.696656227
min	1.255750656
max	39.04676437
dev	6.781374424
median	2.414941788
first_time	39.04676437

--- scenario results (k,n)=(5,9)---

scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	30	3.696656227	1	1
replay_message	30	1.022346814	1	1
wrong_target	30	1.070014636	1	1
mismatched_valu	30	2.115861575	1	1
outdated_ts	30	0.9788274765	1	1

threshold (k,n)	ratio	total node count	avg total time	correct_authorise	register_value_ch	avg_scen_revok	min total time	max total time	median	total time deviatic	first time cost		increasing k	median time
control	0	0	0.6045262019	14.29%	14.29%	0	0.447511673	1.16109848	1.16109848	0.161739717	0.7286071777		(3,30)	1.435399055
(3,5)	0.6	8	2.365255356	100.00%	100.00%	0.7710549566	1.040697098	27.68707275	1.496553421	4.785330253	27.68707275		(5,30)	1.52027607
(3,9)	0.3333333333	12	3.313175837	100.00%	100.00%	1.007421811	1.282930374	35.94779968	1.611590385	6.278436768	35.94779968		(10,30)	2.081871033
(5,9)	0.5555555556	14	3.696656227	100.00%	100.00%	1.005727715	1.255750656	39.04676437	2.414941788	6.781374424	39.04676437		(15,30)	1.758933067
(3,19)	0.1578947368	22	3.003342946	100.00%	100.00%	0.9097152286	1.373529434	32.26041794	1.692771912	5.581285292	32.26041794			
(5,19)	0.2631578947	24	3.166325887	100.00%	100.00%	0.9421520763	1.368284225	36.32545471	1.858830452	6.306699279	36.32545471		increasing n	
(10,19)	0.5263157895	29	4.05523777	100.00%	100.00%	1.057716211	1.471042633	48.48790169	1.666545868	8.618502452	48.48790169		(3,5)	1.496553421
(3,30)	0.1	33	2.962152163	100.00%	100.00%	0.9361174372	1.181125641	34.19709206	1.435399055	5.981779936	34.19709206		(3,9)	1.611590385
(5,30)	0.1666666667	35	2.971076965	100.00%	100.00%	1.084362136	1.167058945	36.69500351	1.52027607	6.398957272	36.69500351		(3,19)	1.692771912
(10,30)	0.3333333333	40	4.718780518	100.00%	100.00%	1.104178694	1.594543457	57.37471581	2.081871033	10.05863282	57.37471581		(3,30)	1.435399055
(12,30)	0.4	42	3.722381592	100.00%	100.00%	1.013023323	1.516103745	59.17096138	1.675009727	10.47629495	59.17096138			
(15,30)	0.5	45	4.454930623	100.00%	100.00%	1.142715083	1.573085785	68.15695763	1.758933067	12.06989683	68.15695763			
average			3.493574171			0.997653152	1.347650181	43.21364923	1.746611162	7.576108208	43.21364923		correct auth rate	correct value cha
													100.00%	100.00%
scenario	(3,5)	(3,9)	(5,9)	(3,19)	(5,19)	(10,19)	(3,30)	(5,30)	(10,30)	(12,30)	(15,30)	avg	100.00%	100.00%
legitimate comma	2.365255356	3.313175837	3.696656227	3.003342946	3.166325887	4.05523777	2.962152163	2.971076965	4.718780518	3.722381592	4.454930623	3.493574171	100.00%	100.00%
replay attack	0.8390903473	1.143980026	1.022346814	1.013708115	1.049939791	1.215171814	1.0242939	1.274053256	1.314727465	1.115544637	1.430980364	1.131257866	100.00%	100.00%
mismatched targ	0.6147782008	0.761961937	1.070014636	0.7830301921	0.7664124171	0.911506017	0.8143424988	0.9254773458	0.9370247523	0.7716576258	0.8845011393	0.8400642511	100.00%	100.00%
mismatched valu	1.504516602	2.093362808	2.115861575	1.832056046	1.837142309	2.01672713	1.982148488	2.321449916	2.140585581	2.199141184	2.113318443	2.014210007	100.00%	100.00%
outdated ts	0.9429057439	1.047889392	0.9788274765	1.029435794	1.107223829	1.242280006	1.106007894	1.126273473	1.239593824	1.203227043	1.435303688	1.132633469	100.00%	100.00%
missing shares	0.1337448756	0.2086321513	0.06872812907	0.05483627319	0.0795284907	0.1408179601	0.0550587972	0.07695357005	0.1209338506	0.1397530238	0.1879851023	0.1151792931		
authorisation skip	0.5912939707	0.7887045542	0.7785876592	0.7452249527	0.8126656214	0.819794337	0.6348530451	0.7819652557	0.8722066879	0.6488164266	0.8042017619	0.7525740248		
stage	control	(3,5)	(3,9)	(5,9)	(3,19)	(5,19)	(10,19)	(3,30)	(5,30)	(10,30)	(12,30)	(15,30)	avg	
make message	0	0.02863407135	0.03388722738	0.04719893138	0.0373840332	0.03568331401	0.0368197759	0.0340461731	0.0340461731	0.04784266154	0.03801186879	0.03605683645	0.03723736965	
share collection	0	0.3776947657	0.5026817322	0.5657196045	0.4659016927	0.5534331004	0.6670792898	0.4978100459	0.4855155945	0.7994890213	0.9924093882	0.9806156158	0.6262136228	
key reconstructio	0	0.02058347066	0.02398490906	0.04436969757	0.0245253245	0.0390291214	0.1093387604	0.02358754476	0.03661314646	0.1123905182	0.142566363	0.1971801122	0.07037899711	
CMAC constructi	0	0.17897288	0.2068916957	0.3789504369	0.2050320307	0.2035856247	0.4344860713	0.2015193303	0.2513647079	0.2482414246	0.2316792806	0.2716700236	0.2556721369	
Key Zeroising	0	0.001875559489	0.002272923787	0.003012021383	0.002225240072	0.002400080363	0.002368291219	0.002121925354	0.002153714498	0.003616015116	0.002026557922	0.002217292786	0.002389665635	
Authentication M	0	1.079201698	1.52156353	1.674032211	1.338966688	1.49658521	1.727747917	1.42103831	1.366567612	2.019715309	1.60039266	2.032558123	1.570760843	
Modbus Instructi	0.6029129028	0.6558338801	0.9890317917	0.9483734767	0.901889801	0.802342097	1.049168905	0.7559855779	0.7676919301	1.448647181	0.6895065308	0.9042183558	0.9011535934	
	(3,5)	(3,9)	(5,9)	(3,19)	(5,19)	(10,19)	(3,30)	(5,30)	(10,30)	(12,30)	(15,30)			
ratio	0.6	0.3333333333	0.5555555556	0.1578947368	0.2631578947	0.5263157895	0.1	0.1666666667	0.3333333333	0.4	0.5			
node count	8	12	14	22	24	29	33	35	40	42	45			
replay attack	0.8390903473	1.143980026	1.022346814	1.013708115	1.049939791	1.215171814	1.0242939	1.274053256	1.314727465	1.115544637	1.430980364			
mismatched targ	0.6147782008	0.761961937	1.070014636	0.7830301921	0.7664124171	0.911506017	0.8143424988	0.9254773458	0.9370247523	0.7716576258	0.8845011393			
mismatched valu	1.504516602	2.093362808	2.115861575	1.832056046	1.837142309	2.01672713	1.982148488	2.321449916	2.140585581	2.199141184	2.113318443			
outdated ts	0.9429057439	1.047889392	0.9788274765	1.029435794	1.107223829	1.242280006	1.106007894	1.126273473	1.239593824	1.203227043	1.435303688			
missing shares	0.1337448756	0.2086321513	0.06872812907	0.05483627319	0.0795284907	0.1408179601	0.0550587972	0.07695357005	0.1209338506	0.1397530238	0.1879851023			
authorisation skip	0.5912939707	0.7887045542	0.7785876592	0.7452249527	0.8126656214	0.819794337	0.6348530451	0.7819652557	0.8722066879	0.6488164266	0.8042017619			



missing_shares	30	0.06872812907	1	1
authorisation_ski	30	0.7785876592	1	1
--- stage timing averages (ms) (k,n)=(3,19) ---				
stage	avg_ms			
make_message	0.0373840332			
total_auth_time	2.052807808			
share_collect_tim	0.4659016927			
key_construction	0.0245253245			
message_signatu	0.2050320307			
zeroise_time	0.002225240072			
auth_time	1.338966688			
share_cost	0.6967782974			
send_message	0.901889801			
total_time	3.003342946			
min	1.373529434			
max	32.26041794			
dev	5.581285292			
median	1.692771912			
first_time	32.26041794			
--- scenario results (k,n)=(3,19)---				
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	30	3.003342946	1	1
replay_message	30	1.013708115	1	1
wrong_target	30	0.7830301921	1	1
mismatched_valu	30	1.832056046	1	1
outdated_ts	30	1.029435794	1	1
missing_shares	30	0.05483627319	1	1
authorisation_ski	30	0.7452249527	1	1
--- stage timing averages (ms) (k,n)=(5,19) ---				
stage	avg_ms			
make_message	0.03568331401			
total_auth_time	2.313168844			
share_collect_tim	0.5534331004			
key_construction	0.0390291214			
message_signatu	0.2035856247			
zeroise_time	0.002400080363			
auth_time	1.49658521			
share_cost	0.7974704107			
send_message	0.802342097			
total_time	3.166325887			
min	1.368284225			
max	36.32545471			
dev	6.306699279			
median	1.858830452			
first_time	36.32545471			
--- scenario results (k,n)=(5,19)---				
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	30	3.166325887	1	1
replay_message	30	1.049939791	1	1
wrong_target	30	0.7664124171	1	1
mismatched_valu	30	1.837142309	1	1
outdated_ts	30	1.107223829	1	1
missing_shares	30	0.0795284907	1	1
authorisation_ski	30	0.8126656214	1	1
--- stage timing averages (ms) (k,n)=(10,19) ---				
stage	avg_ms			
make_message	0.0368197759			
total_auth_time	2.958067258			
share_collect_tim	0.6670792898			
key_construction	0.1093387604			
message_signatu	0.4344860713			
zeroise_time	0.002368291219			
auth_time	1.727747917			
share_cost	1.212374369			
send_message	1.049168905			
total_time	4.05523777			
min	1.471042633			
max	48.48790169			
dev	8.618502452			
median	1.666545868			
first_time	48.48790169			
--- scenario results (k,n)=(10,19)---				
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	30	4.05523777	1	1
replay_message	30	1.215171814	1	1
wrong_target	30	0.911506017	1	1
mismatched_valu	30	2.01672713	1	1
outdated_ts	30	1.242280006	1	1
missing_shares	30	0.1408179601	1	1

authorisation_ski	30	0.819794337	1	1	
--- stage timing averages (ms) (k,n)=(3,30) ---					
stage	avg_ms				
make_message	0.0340461731				
total_auth_time	2.161765099				
share_collect_tin	0.4978100459				
key_construction	0.02358754476				
message_signatu	0.2015193303				
zeroise_time	0.002121925354				
auth_time	1.42103831				
share_cost	0.7241805394				
send_message	0.7559855779				
total_time	2.962152163				
min	1.181125641				
max	34.19709206				
dev	5.981779936				
median	1.435399055				
first_time	34.19709206				
--- scenario results (k,n)=(3,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	30	2.962152163	1	1	
replay_message	30	1.0242939	1	1	
wrong_target	30	0.8143424988	1	1	
mismatched_valu	30	1.982148488	1	1	
outdated_ts	30	1.106007894	1	1	
missing_shares	30	0.0550587972	1	1	
authorisation_ski	30	0.6348530451	1	1	
--- stage timing averages (ms) (k,n)=(5,30) ---					
stage	avg_ms				
make_message	0.0340461731				
total_auth_time	2.15891997				
share_collect_tin	0.4855155945				
key_construction	0.03661314646				
message_signatu	0.2513647079				
zeroise_time	0.002153714498				
auth_time	1.366567612				
share_cost	0.7750034332				
send_message	0.7676919301				
total_time	2.971076965				
min	1.167058945				
max	36.69500351				
dev	6.398957272				
median	1.52027607				
first_time	36.69500351				
--- scenario results (k,n)=(5,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	30	2.971076965	1	1	
replay_message	30	1.274053256	1	1	
wrong_target	30	0.9254773458	1	1	
mismatched_valu	30	2.321449916	1	1	
outdated_ts	30	1.126273473	1	1	
missing_shares	30	0.07695357005	1	1	
authorisation_ski	30	0.7819652557	1	1	
--- stage timing averages (ms) (k,n)=(10,30) ---					
stage	avg_ms				
make_message	0.04784266154				
total_auth_time	3.205887477				
share_collect_tin	0.7994890213				
key_construction	0.1123905182				
message_signatu	0.2482414246				
zeroise_time	0.003616015116				
auth_time	2.019715309				
share_cost	1.162433624				
send_message	1.448647181				
total_time	4.718780518				
min	1.594543457				
max	57.37471581				
dev	10.05863282				
median	2.081871033				
first_time	57.37471581				
--- scenario results (k,n)=(10,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	30	4.718780518	1	1	
replay_message	30	1.314727465	1	1	
wrong_target	30	0.9370247523	1	1	
mismatched_valu	30	2.140585581	1	1	
outdated_ts	30	1.239593824	1	1	
missing_shares	30	0.1209338506	1	1	
authorisation_ski	30	0.8722066879	1	1	

--- stage timing averages (ms) (k,n)=(12,30) ---

stage	avg_ms
make_message	0.03801186879
total_auth_time	2.984635035
share_collect_tim	0.9924093882
key_construction	0.142566363
message_signatu	0.2316792806
zeroise_time	0.002026557922
auth_time	1.60039266
share_cost	1.368014018
send_message	0.6895065308
total_time	3.722381592
min	1.516103745
max	59.17096138
dev	10.47629495
median	1.675009727
first_time	59.17096138

--- scenario results (k,n)=(12,30)---

scenario	count	avg_time_ms	correct_authorisa	register_value_change_rate_%
legitimate_comm	30	3.722381592	1	1
replay_message	30	1.115544637	1	1
wrong_target	30	0.7716576258	1	1
mismatched_valu	30	2.199141184	1	1
outdated_ts	30	1.203227043	1	1
missing_shares	30	0.1397530238	1	1
authorisation_ski	30	0.6488164266	1	1

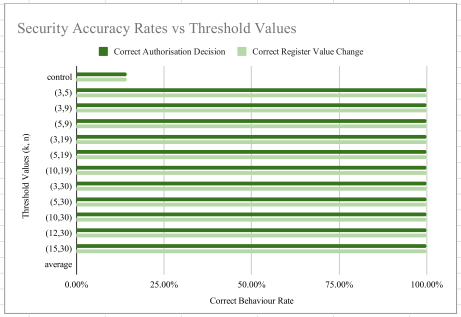
--- stage timing averages (ms) (k,n)=(15,30) ---

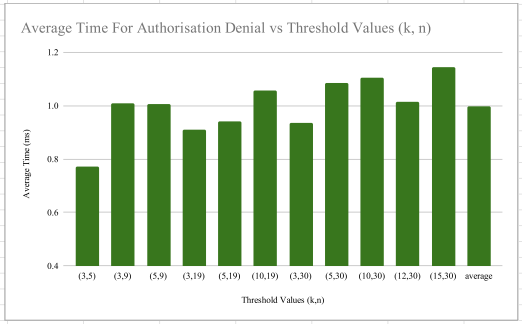
stage	avg_ms
make_message	0.03605683645
total_auth_time	3.502392769
share_collect_tim	0.9806156158
key_construction	0.1971801122
message_signatu	0.2716700236
zeroise_time	0.002217292786
auth_time	2.032558123
share_cost	1.450920105
send_message	0.9042183558
total_time	4.454930623
min	1.573085785
max	68.15695763
dev	12.06989683
median	1.758933067
first_time	68.15695763

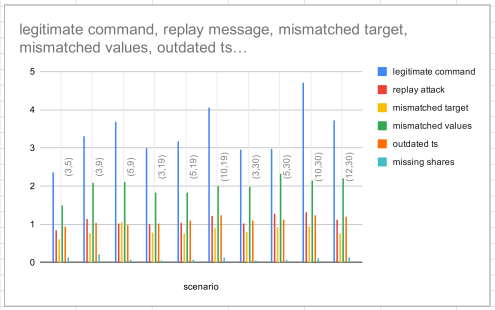
--- scenario results (k,n)=(15,30)---

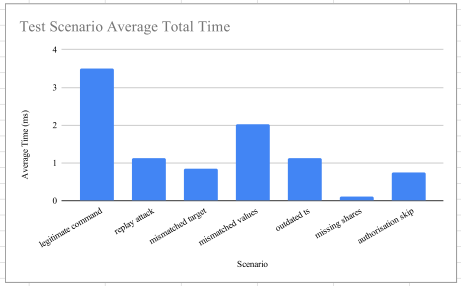
scenario	count	avg_time_ms	correct_authorisa	register_value_change_rate_%
legitimate_comm	30	4.454930623	1	1
replay_message	30	1.430980364	1	1
wrong_target	30	0.8845011393	1	1
mismatched_valu	30	2.113318443	1	1
outdated_ts	30	1.435303688	1	1
missing_shares	30	0.1879851023	1	1
authorisation_ski	30	0.8042017619	1	1

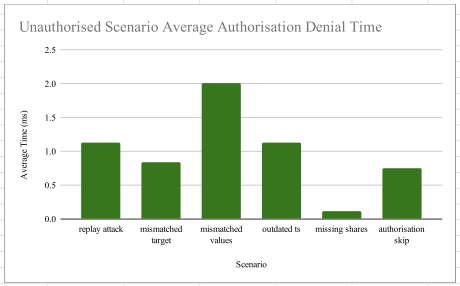
first only (ran with trial count = 1)

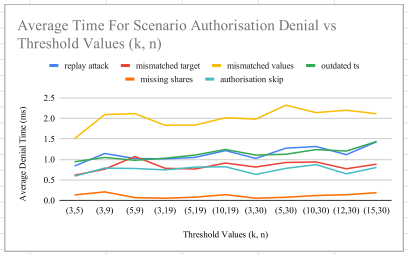

Security Accuracy Rates vs Threshold Values
Correct Authorisation Decision
Correct Register Value Change
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
average
Threshold Values (k, n)
0.00%
25.00%
50.00%
75.00%
100.00%
Correct Behaviour Rate

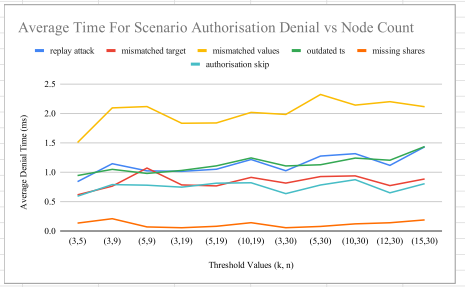

Average Time For Authorisation Denial vs Threshold Values (k, n)
Average Time (ms)
1.2
1.0
0.8
0.6
0.4
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
average
Threshold Values (k,n)

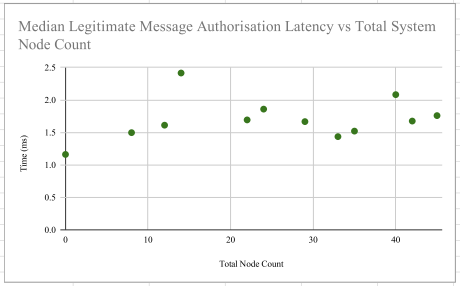

legitimate command, replay message, mismatched target, mismatched values, outdated ts…
legitimate command
replay attack
mismatched target
mismatched values
outdated ts
missing shares
scenario

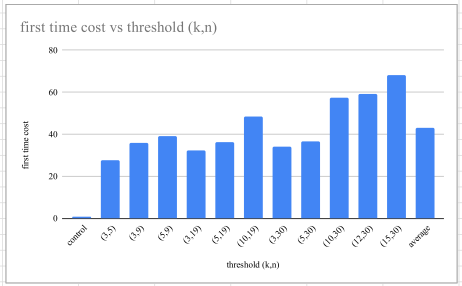

Test Scenario Average Total Time
Average Time (ms)
legitimate command
replay attack
mismatched target
mismatched values
outdated ts
missing shares
authorisation skip
Scenario

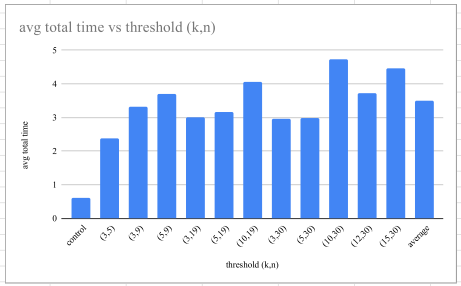

Unauthorised Scenario Average Authorisation Denial Time
Average Time (ms)
replay attack
mismatched target
mismatched values
outdated ts
missing shares
authorisation skip
Scenario

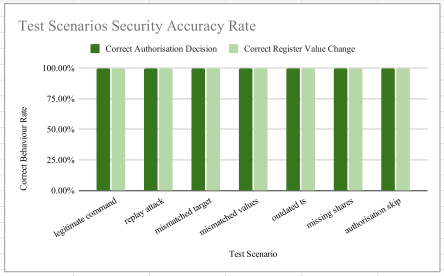

Average Time For Scenario Authorisation Denial vs Threshold Values (k, n)
replay attack
mismatched target
mismatched values
outdated ts
missing shares
authorisation skip
Average Denial Time (ms)
Threshold Values (k, n)

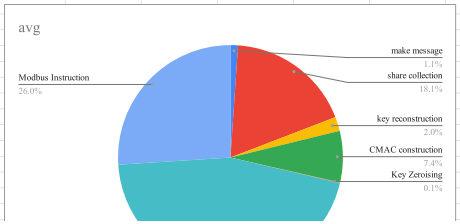

Average Time For Scenario Authorisation Denial vs Node Count
replay attack
mismatched target
mismatched values
outdated ts
missing shares
authorisation skip
Average Denial Time (ms)
Threshold Values (k, n)

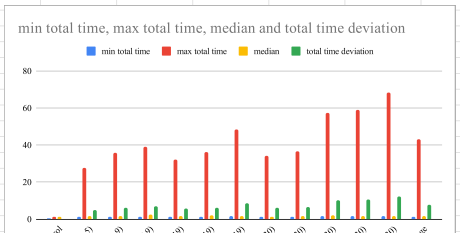

Median Legitimate Message Authorisation Latency vs Total System Node Count
Time (ms)
Total Node Count

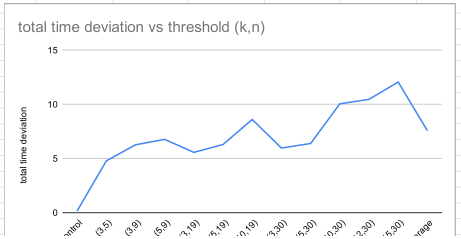

first time cost vs threshold (k,n)
first time cost
threshold (k,n)

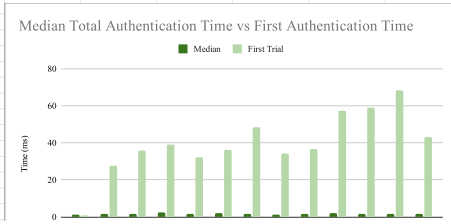

avg total time vs threshold (k,n)
avg total time
threshold (k,n)


Test Scenarios Security Accuracy Rate
Correct Authorisation Decision
Correct Register Value Change
Correct Behaviour Rate
legitimate command
replay attack
mismatched target
mismatched values
outdated ts
missing shares
authorisation skip
Test Scenario

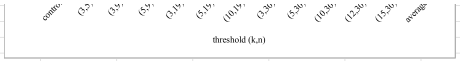

avg
make message
1.1%
share collection
18.1%
Modbus Instruction
26.0%
key reconstruction
2.0%
CMAC construction
7.4%
Key Zeroising
0.1%

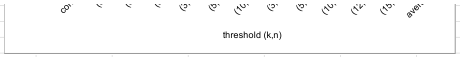

min total time, max total time, median and total time deviation
min total time
max total time
median
total time deviation

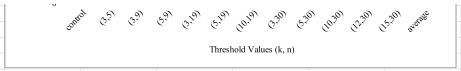

total time deviation vs threshold (k,n)
total time deviation

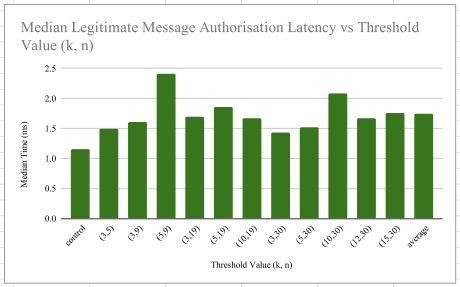

Median Total Authentication Time vs First Authentication Time
Median
First Trial
Time (ms)

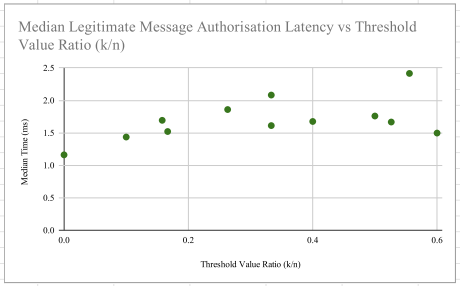

Authentication Message
45.3%

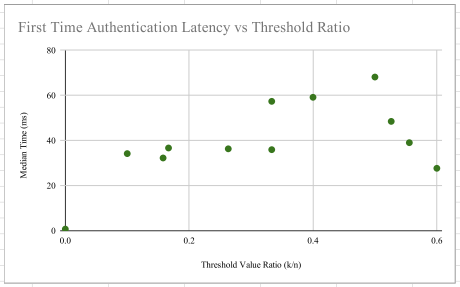

threshold (k,n)

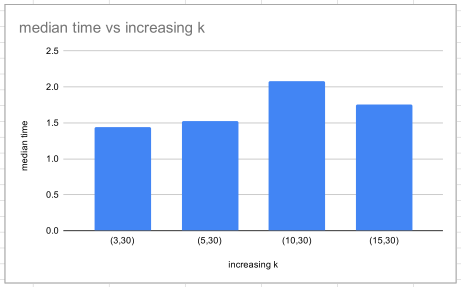

threshold (k,n)

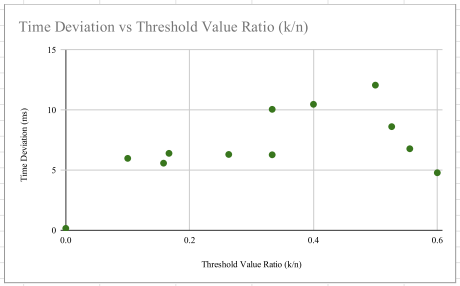

control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
average
Threshold Values (k, n)

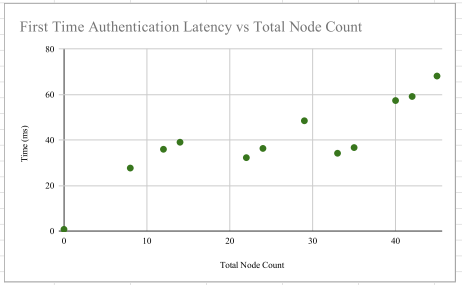

Median Legitimate Message Authorisation Latency vs Threshold Value (k, n)
Median Time (ms)
2.5
2.0
1.5
1.0
0.5
0.0
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
average
Threshold Value (k, n)

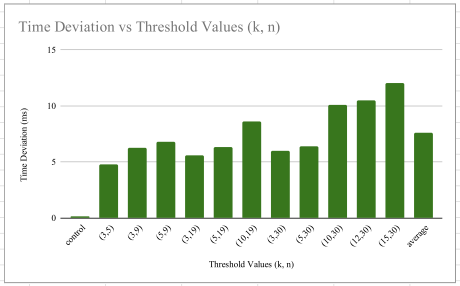

Median Legitimate Message Authorisation Latency vs Threshold Value Ratio (k/n)
Median Time (ms)
2.5
2.0
1.5
1.0
0.5
0.0
0.0
0.2
0.4
0.6
Threshold Value Ratio (k/n)

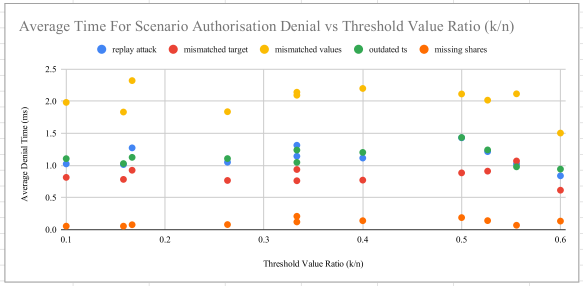

First Time Authentication Latency vs Threshold Ratio
Median Time (ms)
80
60
40
20
0
0.0
0.2
0.4
0.6
Threshold Value Ratio (k/n)

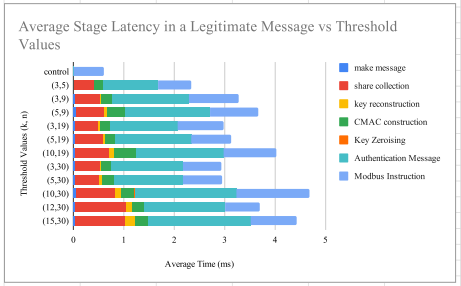

median time vs increasing k
median time
2.5
2.0
1.5
1.0
0.5
0.0
(3,30)
(5,30)
(10,30)
(15,30)
increasing k

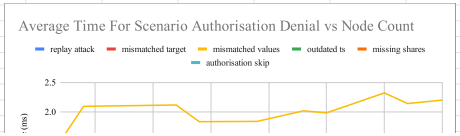

Time Deviation vs Threshold Value Ratio (k/n)
Time Deviation (ms)
15
10
5
0
0.0
0.2
0.4
0.6
Threshold Value Ratio (k/n)

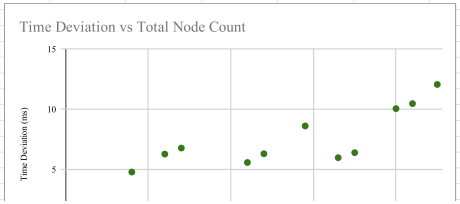

First Time Authentication Latency vs Total Node Count
Time (ms)
80
60
40
20
0
0
10
20
30
40
Total Node Count

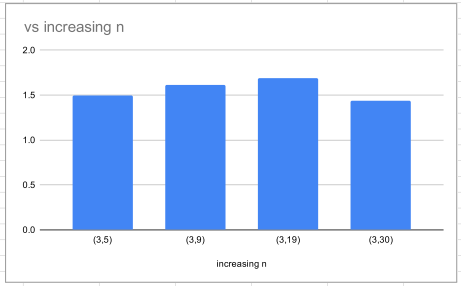

Time Deviation vs Threshold Values (k, n)
Time Deviation (ms)
15
10
5
0
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
average
Threshold Values (k, n)

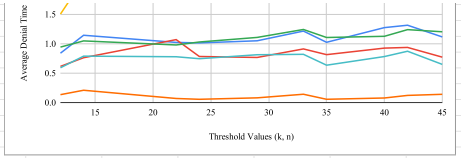

Average Time For Scenario Authorisation Denial vs Threshold Value Ratio (k/n)
replay attack
mismatched target
mismatched values
outdated ts
missing shares
Average Denial Time (ms)
2.5
2.0
1.5
1.0
0.5
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Threshold Value Ratio (k/n)

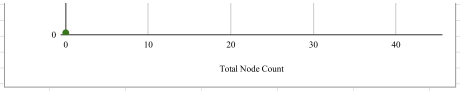

Average Stage Latency in a Legitimate Message vs Threshold Values
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
Threshold Values (k, n)
0
1
2
3
4
5
Average Time (ms)
make message
share collection
key reconstruction
CMAC construction
Key Zeroising
Authentication Message
Modbus Instruction


Average Time For Scenario Authorisation Denial vs Node Count
replay attack
mismatched target
mismatched values
outdated ts
missing shares
authorisation skip
2.5
2.0


Time Deviation vs Total Node Count
Time Deviation (ms)
15
10
5


vs increasing n
2.0
1.5
1.0
0.5
0.0
(3,5)
(3,9)
(3,19)
(3,30)
increasing n


Average Denial Time
1.5
1.0
0.5
0.0
15
20
25
30
35
40
45
Threshold Values (k, n)


0
0
10
20
30
40
Total Node Count

| | | |

--- scenario results (k,n)=(5,9)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	1	30.6994915	1	1	
replay_message	1	0.8709430695	1	1	
wrong_target	1	0.5950927734	1	1	
mismatched_valu	1	1.428842545	1	1	
outdated_ts	1	0.8502006531	1	1	
missing_shares	1	0.06127357483	1	1	
authorisation_ski	1	0.529050827	1	1	
--- stage timing averages (ms) (k,n)=(3,19) ---					
stage	avg_ms				
make_message	0.03528594971				
total_auth_time	37.85276413				
share_collect_tim	15.57159424				
key_construction	0.0319480896				
message_signatu	2.156734467				
zeroise_time	0.003099441528				
auth_time	20.05720139				
share_cost	17.76480675				
send_message	2.568483353				
total_time	40.48871994				
min	40.48871994				
max	40.48871994				
dev	0				
median	40.48871994				
first_time	40.48871994				
--- scenario results (k,n)=(3,19)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	1	40.48871994	1	1	
replay_message	1	0.9400844574	1	1	
wrong_target	1	0.7009506226	1	1	
mismatched_valu	1	1.524925232	1	1	
outdated_ts	1	0.9422302246	1	1	
missing_shares	1	0.1020431519	1	1	
authorisation_ski	1	0.4136562347	1	1	
--- stage timing averages (ms) (k,n)=(5,19) ---					
stage	avg_ms				
make_message	0.0331401825				
total_auth_time	38.18941116				
share_collect_tim	14.75811005				
key_construction	0.05626678467				
message_signatu	2.002954483				
zeroise_time	0.009536743164				
auth_time	21.29745483				
share_cost	16.8223381				
send_message	1.87253952				
total_time	40.13061523				
min	40.13061523				
max	40.13061523				
dev	0				
median	40.13061523				
first_time	40.13061523				
--- scenario results (k,n)=(5,19)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	1	40.13061523	1	1	
replay_message	1	1.158237457	1	1	
wrong_target	1	0.8194446564	1	1	
mismatched_valu	1	1.890420914	1	1	
outdated_ts	1	2.09736824	1	1	
missing_shares	1	0.1137256622	1	1	
authorisation_ski	1	0.7786750793	1	1	
--- stage timing averages (ms) (k,n)=(10,19) ---					
stage	avg_ms				
make_message	0.03385543823				
total_auth_time	43.68495941				
share_collect_tim	17.619133				
key_construction	0.1549720764				
message_signatu	1.765966415				
zeroise_time	0.003814697266				
auth_time	24.10936356				
share_cost	19.54483986				
send_message	1.628160477				
total_time	45.36271095				

min	45.36271095				
max	45.36271095				
dev	0				
median	45.36271095				
first_time	45.36271095				
--- scenario results (k,n)=(10,19)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	1	45.36271095	1	1	
replay_message	1	0.9248256683	1	1	
wrong_target	1	0.7219314575	1	1	
mismatched_valu	1	1.580715179	1	1	
outdated_ts	1	0.953912735	1	1	
missing_shares	1	0.1373291016	1	1	
authorisation_ski	1	1.639127731	1	1	
--- stage timing averages (ms) (k,n)=(3,30) ---					
stage	avg_ms				
make_message	0.03051757813				
total_auth_time	30.53832054				
share_collect_tim	11.70229912				
key_construction	0.02336502075				
message_signatu	1.091718674				
zeroise_time	0.002861022949				
auth_time	17.69351959				
share_cost	12.82024384				
send_message	0.9250640869				
total_time	31.51321411				
min	31.51321411				
max	31.51321411				
dev	0				
median	31.51321411				
first_time	31.51321411				
--- scenario results (k,n)=(3,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	1	31.51321411	1	1	
replay_message	1	2.438545227	1	1	
wrong_target	1	2.114057541	1	1	
mismatched_valu	1	3.739356995	1	1	
outdated_ts	1	2.335309982	1	1	
missing_shares	1	0.1277923584	1	1	
authorisation_ski	1	1.246213913	1	1	
--- stage timing averages (ms) (k,n)=(5,30) ---					
stage	avg_ms				
make_message	0.0376701355				
total_auth_time	34.46555138				
share_collect_tim	14.80579376				
key_construction	0.03814697266				
message_signatu	1.255512238				
zeroise_time	0.004291534424				
auth_time	18.33319664				
share_cost	16.10302925				
send_message	1.02853775				
total_time	35.54868698				
min	35.54868698				
max	35.54868698				
dev	0				
median	35.54868698				
first_time	35.54868698				
--- scenario results (k,n)=(5,30)---					
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%	
legitimate_comm	1	35.54868698	1	1	
replay_message	1	2.044439316	1	1	
wrong_target	1	0.9391307831	1	1	
mismatched_valu	1	2.043008804	1	1	
outdated_ts	1	1.239061356	1	1	
missing_shares	1	0.09107589722	1	1	
authorisation_ski	1	0.5934238434	1	1	
--- stage timing averages (ms) (k,n)=(10,30) ---					
stage	avg_ms				
make_message	0.03504753113				
total_auth_time	42.709589				
share_collect_tim	17.13776588				
key_construction	0.1125335693				

message_signatu	1.070737839																								
zeroise_time	0.003814697266																								
auth_time	24.36041832																								
share_cost	18.32413673																								
send_message	0.9229183197																								
total_time	43.68710518																								
min	43.68710518																								
max	43.68710518																								
dev	0																								
median	43.68710518																								
first_time	43.68710518																								
--- scenario results (k,n)=(10,30)---																									
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%																					
legitimate_comm	1	43.68710518	1	1																					
replay_message	1	1.441717148	1	1																					
wrong_target	1	0.7567405701	1	1																					
mismatched_valu	1	2.424240112	1	1																					
outdated_ts	1	1.229524612	1	1																					
missing_shares	1	0.1525878906	1	1																					
authorisation_ski	1	0.7996559143	1	1																					
--- stage timing averages (ms) (k,n)=(12,30) ---																									
stage	avg_ms																								
make_message	0.02908706665																								
total_auth_time	43.15185547																								
share_collect_tim	19.7789669																								
key_construction	0.1454353333																								
message_signatu	1.111030579																								
zeroise_time	0.003099441528																								
auth_time	22.08900452																								
share_cost	21.037817																								
send_message	1.037120819																								
total_time	44.23761368																								
min	44.23761368																								
max	44.23761368																								
dev	0																								
median	44.23761368																								
first_time	44.23761368																								
--- scenario results (k,n)=(12,30)---																									
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%																					
legitimate_comm	1	44.23761368	1	1																					
replay_message	1	1.018047333	1	1																					
wrong_target	1	0.7257461548	1	1																					
mismatched_valu	1	1.63435936	1	1																					
outdated_ts	1	0.9732246399	1	1																					
missing_shares	1	0.1537799835	1	1																					
authorisation_ski	1	0.6783008575	1	1																					
--- stage timing averages (ms) (k,n)=(15,30) ---																									
stage	avg_ms																								
make_message	0.03838539124																								
total_auth_time	61.89751625																								
share_collect_tim	27.8980732																								
key_construction	0.1540184021																								
message_signatu	1.128435135																								
zeroise_time	0.002384185791																								
auth_time	32.69219398																								
share_cost	29.18314934																								
send_message	0.9391307831																								
total_time	62.89148331																								
min	62.89148331																								
max	62.89148331																								
dev	0																								
median	62.89148331																								
first_time	62.89148331																								
--- scenario results (k,n)=(15,30)---																									
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%																					
legitimate_comm	1	62.89148331	1	1																					
replay_message	1	1.074314117	1	1																					
wrong_target	1	0.7631778717	1	1																					
mismatched_valu	1	1.657962799	1	1																					
outdated_ts	1	1.122951508	1	1																					
missing_shares	1	0.1938343048	1	1																					
authorisation_ski	1	0.8120536804	1	1																					

--- control timing results ---										
cnt	avg_time	avg_send_msg_t	first_time	min	max	dev	median	correct_authorise	correct_register_value_change_%	
29	0.6002475475	0.5986608308	0.6573200226	0.447511673	1.16109848	0.1654770611	0.5421638489	1	1	
--- stage timing averages (ms) (k,n)=(3,5) ---										
stage	avg_ms									
make_message	0.02837181091									
total_auth_time	0.8111164488									
share_collect_tim	0.001397626153									
key_construction	0.02035601386									
message_signatu	0.1466849755									
zeroise_time	0.001866241981									
auth_time	0.6274765935									
share_cost	0.1694827244									
send_message	0.6439521395									
total_time	1.492089239									
min	1.040697098									
max	1.973390579									
dev	0.1694001553									
median	1.486778259									
--- scenario results (k,n)=(3,5)---										
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%						
legitimate_comm	29	1.492089239	1	1						
replay_message	29	0.8358955383	1	1						
wrong_target	29	0.6134345614	1	1						
mismatched_valu	29	1.502579656	1	1						
outdated_ts	29	0.944754173	1	1						
missing_shares	29	0.1366220672	1	1						
authorisation_ski	29	0.5904066152	1	1						
--- stage timing averages (ms) (k,n)=(3,9) ---										
stage	avg_ms									
make_message	0.0340363075									
total_auth_time	1.145658822									
share_collect_tim	0.001652487393									
key_construction	0.02302794621									
message_signatu	0.1752787623									
zeroise_time	0.002277308497									
auth_time	0.9245461431									
share_cost	0.2012499448									
send_message	0.9945589921									
total_time	2.18784398									
min	1.282930374									
max	6.743431091									
dev	1.227479464									
median	1.554012299									
--- scenario results (k,n)=(3,9)---										
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%						
legitimate_comm	29	2.18784398	1	1						
replay_message	29	1.12321459	1	1						
wrong_target	29	0.7618789015	1	1						
mismatched_valu	29	2.076790251	1	1						
outdated_ts	29	1.053818341	1	1						
missing_shares	29	0.2140587774	1	1						
authorisation_ski	29	0.7962522836	1	1						
--- stage timing averages (ms) (k,n)=(5,9) ---										
stage	avg_ms									
make_message	0.04732197729									
total_auth_time	1.479362619									
share_collect_tim	0.00235130047									
key_construction	0.0448309142									
message_signatu	0.3625277815									
zeroise_time	0.003041892216									
auth_time	1.046509578									
share_cost	0.4113049343									
send_message	0.9364177441									
total_time	2.477686981									
min	1.255750656									
max	6.24871254									
dev	1.230813798									
median	2.189040184									
--- scenario results (k,n)=(5,9)---										
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%						
legitimate_comm	29	2.477686981	1	1						
replay_message	29	1.030790395	1	1						
wrong_target	29	1.079879958	1	1						
mismatched_valu	29	2.112454381	1	1						

threshold (k,n)	ratio	node count	avg total time	correct_authorise	register_value_cl	avg_scen_revok	min total time	max total time	median	time deviation		element	avg time (ms)
control	0.00%	0	0.6002475475	14.29%	14.29%	0	0.447511673	1.16109848	0.5421638489	0.1654770611		avg total time	2.164638902
(3,5)	60.00%	8	1.492089239	100.00%	100.00%	0.8736831214	1.040697098	1.973390579	1.486778259	0.1694001553		avg_scen_revoke	1.018875943
(3,9)	33.33%	12	2.18784398	100.00%	100.00%	1.004335524	1.282930374	6.743431091	1.554012299	1.227479464		min time	1.347650181
(5,9)	55.56%	14	2.477686981	100.00%	100.00%	1.009153224	1.255750656	6.24871254	2.189040184	1.230813798		max total time	5.069450899
(3,19)	15.79%	22	1.994478291	100.00%	100.00%	1.055230061	1.373529434	4.959821701	1.630783081	0.8129931529		median	1.705202189
(5,19)	26.32%	24	2.022907652	100.00%	100.00%	0.9324427309	1.368284225	4.016876221	1.830935478	0.7701756518		deviation	0.966527132
(10,19)	52.63%	29	2.523076945	100.00%	100.00%	1.054954255	1.471042633	9.884357452	1.64437294	2.028528348			
(3,30)	10.00%	33	1.88508527	100.00%	100.00%	0.9371987705	1.181125641	4.427909851	1.432061195	0.9197514475			
(5,30)	16.67%	35	1.808182947	100.00%	100.00%	1.094201515	1.167058945	3.751039505	1.502037048	0.6340737647			
(10,30)	33.33%	40	2.903058611	100.00%	100.00%	1.08910292	1.594543457	6.16645813	2.067804337	1.554977555			
(12,30)	0.4	42	2.258309003	100.00%	100.00%	1.01722383	1.516103745	2.625465393	1.673340797	0.2894805537			
(15,30)	50.00%	45	2.258309003	100.00%	100.00%	1.140109424	1.573085785	4.966497421	1.746058464	0.9941245618			

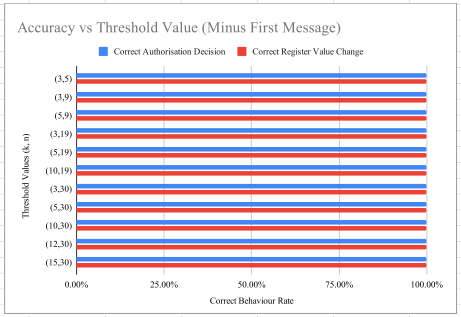
scenario	(3,5)	(3,9)	(5,9)	(3,19)	(5,19)	(10,19)	(3,30)	(5,30)	(10,30)	(12,30)	(15,30)	avg
legitimate_comm	1.492089239	2.18784398	2.477686981	1.994478291	2.022907652	2.523076945	1.88508527	1.808182947	2.903058611	1.810361599	2.258309003	2.123916411
replay_message	0.8358955383	1.12321459	1.030790395	0.9891740207	1.058405843	1.212744877	0.9985463373	1.288578428	1.226260744	1.120254911	1.442465289	1.120575543
wrong_target	0.6134345614	0.7618789015	1.079879958	0.7816183156	0.7720815724	0.9164070261	0.8165836334	0.9323070789	0.9450419196	0.7736189612	0.8872048608	0.8436415263
mismatched_valu	1.502579656	2.076790251	2.112454381	1.840040602	1.794132693	2.023811998	2.000471641	2.339083573	2.156685139	2.215007256	2.125041238	2.016918039
outdated_ts	0.944754173	1.053818341	0.9752799725	0.9760774415	1.068000136	1.241593525	1.115174129	1.138136305	1.243295341	1.204474219	1.384200721	1.122254937
missing_shares	0.1366220672	0.2140587774	0.06901806798	0.05509935576	0.08017441322	0.1417110706	0.05531311035	0.07718184899	0.1204835957	0.139705066	0.1883835628	0.116159176
authorisation_ski	0.5904066152	0.7962522836	0.7874965668	0.7501223992	0.8218617275	0.7934570313	0.6371037713	0.7899218592	0.8428507838	0.6502825638	0.8133608719	0.7521014976

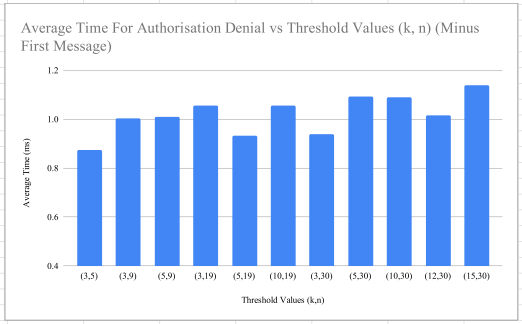
scenario	control	(3,5)	(3,9)	(5,9)	(3,19)	(5,19)	(10,19)	(3,30)	(5,30)	(10,30)	(12,30)	(15,30)	avg
make message	0	0.02837181091	0.0340363075	0.04732197729	0.03741527426	0.03591899214	0.03668357586	0.03425828342	0.03406097149	0.04797146238	0.03020516757	0.03584500017	0.03655352936
share collection	0	0.001397626153	0.001652487393	0.00235130047	0.001611380742	0.001923791293	0.001619602072	0.001751143357	0.001767586017	0.00235130047	0.001446954135	0.001981340606	0.001804955701
key reconstructio	0	0.02035601386	0.02302794621	0.0448309142	0.02412138314	0.03932262289	0.1092532585	0.0235130047	0.03609986141	0.1091134959	0.1410451429	0.1926093266	0.06939027003
CMAC creation	0	0.1466849755	0.1752787623	0.3625277815	0.175147221	0.1800060272	0.3845198401	0.171439401	0.1754596316	0.1922064814	0.150557222	0.1740784481	0.2079914356
key zeroisation	0	0.001866241981	0.002277308497	0.003041892216	0.002227980515	0.002392407121	0.002252644506	0.002079996569	0.002063553909	0.003156990841	0.001940233954	0.001964897945	0.002296740732
authorisation mes	0	0.6274765935	0.9245461431	1.046509578	0.8212862344	0.9256231374	0.9100930444	0.8796083516	0.7719664738	1.148256762	0.78065642	0.9577685389	0.8903446616
Modbus instructio	0.5986608308	0.6439521395	0.9945589921	0.9364177441	0.9055877554	0.8048271311	1.051327278	0.7468174244	0.7609663338	1.36577672	0.6797642543	0.8650154903	0.8868192057

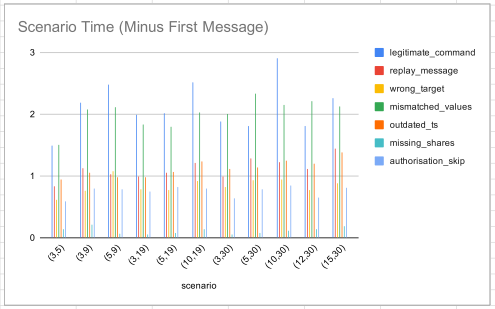
outdated_ts	29	0.9752799725	1	1	
missing_shares	29	0.06901806798	1	1	
authorisation_ski	29	0.7874965668	1	1	
--- stage timing averages (ms) (k,n)=(3,19) ---					
stage	avg_ms				
make_message	0.03741527426				
total_auth_time	1.040360023				
share_collect_tim	0.001611380742				
key_construction	0.02412138314				
message_signatu	0.175147221				
zeroise_time	0.002227980515				
auth_time	0.8212862344				
share_cost	0.2021296271				
send_message	0.9055877554				
total_time	1.994478291				
min	1.373529434				
max	4.959821701				
dev	0.8129931529				
median	1.630783081				
--- scenario results (k,n)=(3,19)---					
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%
legitimate_comm	29	1.994478291	1	1	
replay_message	29	0.9891740207	1	1	
wrong_target	29	0.7816183156	1	1	
mismatched_valu	29	1.840040602	1	1	
outdated_ts	29	0.9760774415	1	1	
missing_shares	29	0.05509935576	1	1	
authorisation_ski	29	0.7501223992	1	1	
--- stage timing averages (ms) (k,n)=(5,19) ---					
stage	avg_ms				
make_message	0.03591899214				
total_auth_time	1.167026059				
share_collect_tim	0.001923791293				
key_construction	0.03932262289				
message_signatu	0.1800060272				
zeroise_time	0.002392407121				
auth_time	0.9256231374				
share_cost	0.2226582889				
send_message	0.8048271311				
total_time	2.022907652				
min	1.368284225				
max	4.016876221				
dev	0.7701756518				
median	1.830935478				
--- scenario results (k,n)=(5,19)---					
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%
legitimate_comm	29	2.022907652	1	1	
replay_message	29	1.058405843	1	1	
wrong_target	29	0.7720815724	1	1	
mismatched_valu	29	1.794132693	1	1	
outdated_ts	29	1.068000136	1	1	
missing_shares	29	0.08017441322	1	1	
authorisation_ski	29	0.8218617275	1	1	
--- stage timing averages (ms) (k,n)=(10,19) ---					
stage	avg_ms				
make_message	0.03668357586				
total_auth_time	1.424000181				
share_collect_tim	0.001619602072				
key_construction	0.1092532585				
message_signatu	0.3845198401				
zeroise_time	0.002252644506				
auth_time	0.9100930444				
share_cost	0.4967656629				
send_message	1.051327278				
total_time	2.523076945				
min	1.471042633				
max	9.884357452				
dev	2.028528348				
median	1.64437294				
--- scenario results (k,n)=(10,19)---					
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%
legitimate_comm	29	2.523076945	1	1	
replay_message	29	1.212744877	1	1	
wrong_target	29	0.9164070261	1	1	
mismatched_valu	29	2.023811998	1	1	

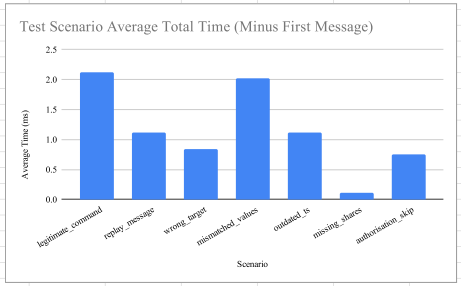
outdated_ts	29	1.241593525	1	1
missing_shares	29	0.1417110706	1	1
authorisation_ski	29	0.7934570313	1	1
--- stage timing averages (ms) (k,n)=(3,30) ---				
stage	avg_ms			
make_message	0.03425828342			
total_auth_time	1.093864441			
share_collect_tim	0.001751143357			
key_construction	0.0235130047			
message_signatu	0.171439401			
zeroise_time	0.002079996569			
auth_time	0.8796083516			
share_cost	0.19794497			
send_message	0.7468174244			
total_time	1.88508527			
min	1.181125641			
max	4.427909851			
dev	0.9197514475			
median	1.432061195			
--- scenario results (k,n)=(3,30)---				
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	29	1.88508527	1	1
replay_message	29	0.9985463373	1	1
wrong_target	29	0.8165836334	1	1
mismatched_valu	29	2.000471641	1	1
outdated_ts	29	1.115174129	1	1
missing_shares	29	0.05531311035	1	1
authorisation_ski	29	0.6371037713	1	1
--- stage timing averages (ms) (k,n)=(5,30) ---				
stage	avg_ms			
make_message	0.03406097149			
total_auth_time	1.002944749			
share_collect_tim	0.001767586017			
key_construction	0.03609986141			
message_signatu	0.1754596316			
zeroise_time	0.002063553909			
auth_time	0.7719664738			
share_cost	0.2146835985			
send_message	0.7609663338			
total_time	1.808182947			
min	1.167058945			
max	3.751039505			
dev	0.6340737647			
median	1.502037048			
--- scenario results (k,n)=(5,30)---				
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	29	1.808182947	1	1
replay_message	29	1.288578428	1	1
wrong_target	29	0.9323070789	1	1
mismatched_valu	29	2.339083573	1	1
outdated_ts	29	1.138136305	1	1
missing_shares	29	0.07718184899	1	1
authorisation_ski	29	0.7899218592	1	1
--- stage timing averages (ms) (k,n)=(10,30) ---				
stage	avg_ms			
make_message	0.04797146238			
total_auth_time	1.474660018			
share_collect_tim	0.00235130047			
key_construction	0.1091134959			
message_signatu	0.1922064814			
zeroise_time	0.003156990841			
auth_time	1.148256762			
share_cost	0.3052908799			
send_message	1.36577672			
total_time	2.903058611			
min	1.594543457			
max	6.16645813			
dev	1.554977555			
median	2.067804337			
--- scenario results (k,n)=(10,30)---				
scenario	count	avg_time_ms	correct_authorise	register_value_change_rate_%
legitimate_comm	29	2.903058611	1	1
replay_message	29	1.226260744	1	1
wrong_target	29	0.9450419196	1	1
mismatched_valu	29	2.156685139	1	1

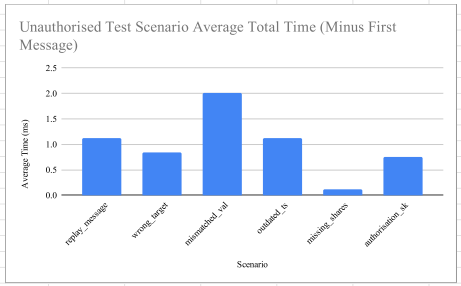
outdated_ts	29	1.243295341	1	1																							
missing_shares	29	0.1204835957	1	1																							
authorisation_ski	29	0.8428507838	1	1																							
--- stage timing averages (ms) (k,n)=(12,30) ---																											
stage	avg_ms																										
make_message	0.03020516757																										
total_auth_time	1.090378597																										
share_collect_tir	0.001446954135																										
key_construction	0.1410451429																										
message_signatu	0.150557222																										
zeroise_time	0.001940233954																										
auth_time	0.78065642																										
share_cost	0.2942660759																										
send_message	0.6797642543																										
total_time	1.810361599																										
min	1.516103745																										
max	2.625465393																										
dev	0.2894805537																										
median	1.673340797																										
--- scenario results (k,n)=(12,30)---																											
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%																						
legitimate_comm	29	1.810361599	1	1																							
replay_message	29	1.120254911	1	1																							
wrong_target	29	0.7736189612	1	1																							
mismatched_valu	29	2.215007256	1	1																							
outdated_ts	29	1.204474219	1	1																							
missing_shares	29	0.139705066	1	1																							
authorisation_ski	29	0.6502825638	1	1																							
--- stage timing averages (ms) (k,n)=(15,30) ---																											
stage	avg_ms																										
make_message	0.03584500017																										
total_auth_time	1.345519362																										
share_collect_tir	0.001981340606																										
key_construction	0.1926093266		8.53%																								
message_signatu	0.1740784481																										
zeroise_time	0.001964897945																										
auth_time	0.9577685389																										
share_cost	0.3698283228																										
send_message	0.8650154903																										
total_time	2.258309003																										
min	1.573085785																										
max	4.966497421																										
dev	0.9941245618																										
median	1.746058464																										
--- scenario results (k,n)=(15,30)---																											
scenario	count	avg_time_ms	correct_authorise	register_value_	change_rate_%																						
legitimate_comm	29	2.258309003	1	1																							
replay_message	29	1.442465289	1	1																							
wrong_target	29	0.8872048608	1	1																							
mismatched_valu	29	2.125041238	1	1																							
outdated_ts	29	1.384200721	1	1																							
missing_shares	29	0.1883835628	1	1																							
authorisation_ski	29	0.8133608719	1	1																							

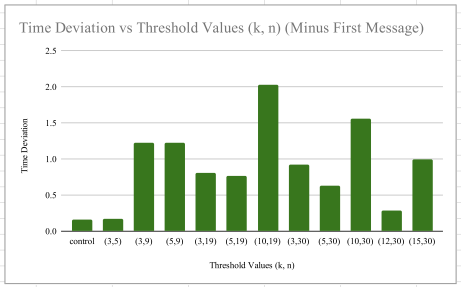

Accuracy vs Threshold Value (Minus First Message)
Correct Authorisation Decision
Correct Register Value Change
Threshold Values (k, n)
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
0.00%
25.00%
50.00%
75.00%
100.00%
Correct Behaviour Rate

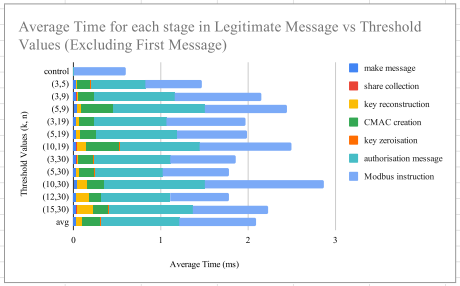

Average Time For Authorisation Denial vs Threshold Values (k, n) (Minus First Message)
Average Time (ms)
1.2
1.0
0.8
0.6
0.4
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
Threshold Values (k,n)

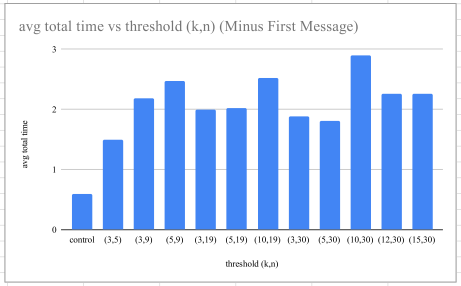

Scenario Time (Minus First Message)
legitimate_command
replay_message
wrong_target
mismatched_values
outdated_ts
missing_shares
authorisation_skip
scenario

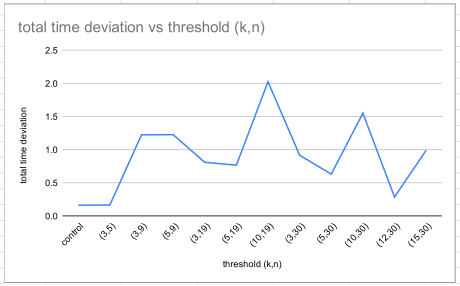

Test Scenario Average Total Time (Minus First Message)
Average Time (ms)
legitimate_command
replay_message
wrong_target
mismatched_values
outdated_ts
missing_shares
authorisation_skip
Scenario

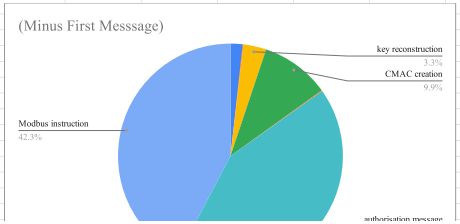

Unauthorised Test Scenario Average Total Time (Minus First Message)
Average Time (ms)
replay_message
wrong_target
mismatched_val
outdated_ts
missing_shares
authorisation_sk
Scenario


Time Deviation vs Threshold Values (k, n) (Minus First Message)
Time Deviation
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
Threshold Values (k, n)

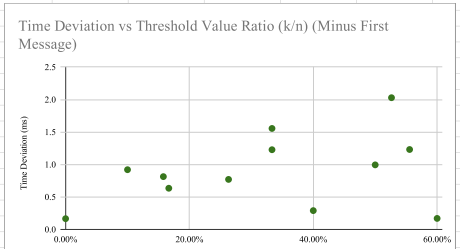

Average Time for each stage in Legitimate Message vs Threshold Values (Excluding First Message)
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
avg
Threshold Values (k, n)
make message
share collection
key reconstruction
CMAC creation
key zeroisation
authorisation message
Modbus instruction
Average Time (ms)

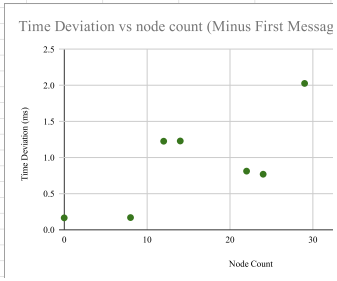

avg total time vs threshold (k,n) (Minus First Message)
avg total time
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
threshold (k,n)

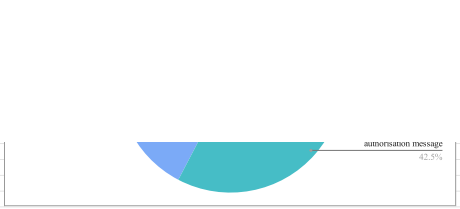

total time deviation vs threshold (k,n)
total time deviation
control
(3,5)
(3,9)
(5,9)
(3,19)
(5,19)
(10,19)
(3,30)
(5,30)
(10,30)
(12,30)
(15,30)
threshold (k,n)

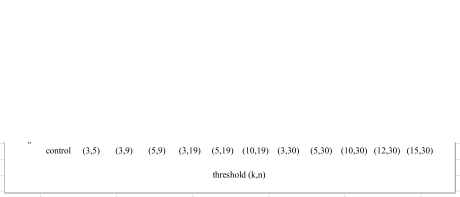

(Minus First Messsage)
key reconstruction
3.3%
CMAC creation
9.9%
Modbus instruction
42.5%
authorisation message

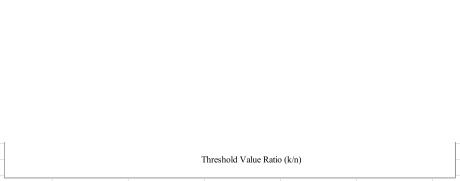

min total time, max total time, median and total time deviation (Minus First Mesage)
min total time
max total time
median
time deviation


Time Deviation vs Threshold Value Ratio (k/n) (Minus First Message)
Time Deviation (ms)
0.00%
20.00%
40.00%
60.00%


Time Deviation vs node count (Minus First Messag
Time Deviation (ms)
Node Count


authorisation message
42.5%


control (3,5) (3,9) (5,9) (3,19) (5,19) (10,19) (3,30) (5,30) (10,30) (12,30) (15,30)
threshold (k,n)


Threshold Value Ratio (k/n)